Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: EET 301

Course Name: POWER SYSTEMS I

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

		Marks
1	List the ethical and environmental factors associated with a nuclear power plant.	3
2	Explain utility scale battery energy storage system.	3
3	Elucidate on the importance of transposition in transmission lines	3
4	Find the capacitance of a single-phase line 40km long consisting of two parallel conductors each 5mm in diameter and 1.5 m apart. The height of conductors above ground is 7m.The effect of ground may be included.	3
5	What are the factors that affecting the phenomenon corona in transmission lines?	3
6	HVDC transmission system is preferred over HVAC transmission system for overhead transmission line lengths above 800 km. Justify the statement.	3
7	Define the terms Restriking voltage, Recovery voltage and rate of rise of restriking voltage with reference to operation of circuit breakers	3
8	List the causes of over voltages in a power system.	3
9	Distinguish between radial and ring main distribution systems	3
10	Elucidate the use of ABC cable in power system	3

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	With a neat block diagram explain the working of a thermal power plant	7
	b)	A diesel station supplies the following loads to various consumers : Industrial consumer = 1500 kW ; Commercial establishment = 750 kW Domestic power = 100 kW; Domestic light = 450 kW. If the maximum demand on the station is 2500 kW and the number of kWh generated per year is 45×10^6, determine (i) the diversity factor and (ii) annual load factor.	7

12 a) Differentiate between rooftop and ground mounted solar power plants 6

b) A generating station has the following daily load cycle : 8

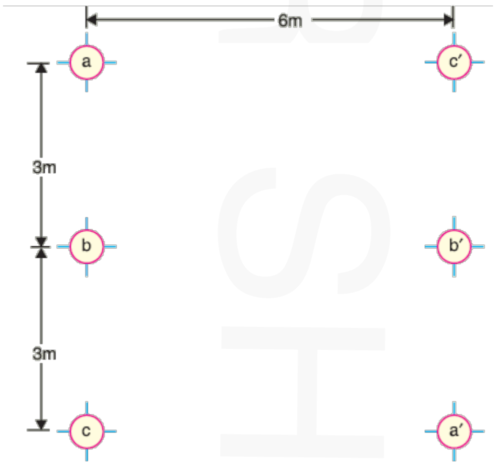
Time (Hours)	0—6	6—10	10—12	12—16	16—20	20—24
Load (MW)	40	50	60	50	70	40

Draw the load curve and find (i) maximum demand (ii) units generated per day (iii) average load and (iv) load factor.

Module -2

13 a) Derive the expression for capacitance of a three phase overhead transmission line assuming symmetrical spacing between conductors 7

b) Figure shows the spacings of a double circuit 3-phase overhead line. The phase sequence is ABC and the line is completely transposed. The conductor radius in 1·3 cm. Find the inductance per phase per kilometre. 7



14 a) Derive the expression for sending end voltage and sending end current in terms of receiving end voltage and receiving end current for long transmission line. 6

b) A 3-phase, 50-Hz overhead transmission line 100 km long has the following constants : 8

Resistance/km/phase = 0.1 Ω

Inductive reactance/km/phase = 0·2 Ω

Capacitive susceptance/km/phase = $0{\cdot}04 \times 10^{-4}$ siemen

Determine (i) the sending end current (ii) sending end voltage (iii) sending end power factor and (iv) transmission efficiency when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0·8 lagging. Use nominal T method.

Module -3

15 a) The capacitances of a 3-phase belted cable are 12·6 μF between the three cores bunched together and the lead sheath and 7·4 μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. 8

b) Explain the methods of improving string efficiency 6

16 a) An overhead transmission line conductor having a parabolic configuration weighs 1·925 kg per metre of length. The area of X-section of the conductor is 2·2 cm^2 and the ultimate strength is 8000 kg/cm^2. The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that ice load is 1 kg per metre run and there is no wind pressure. 6

b) Explain the significance of Flexible AC Transmission System (FACTS) devices in power system. Explain the working of any two FACTS devices. List out the demerits of FACTS devices. 8

Module -4

17 a) Explain the term duality in terms of amplitude and phase comparators in static relays 6

b) With the help of a neat diagram explain the working of SF_6 circuit breaker. List the advantages of SF_6 when used in Gas Insulated substation. 8

18 a) Draw and explain the block diagram of microprocessor based over current relay 5

b) Explain with a neat circuit diagram the working principle of differential relay. 5

c) Explain the principle of fibre optic communication in transmission system 4

Module -5

19 a) A single phase a.c. generator supplies the following loads : 7

(i) Lighting load of 20 kW at unity power factor.

(ii) Induction motor load of 100 kW at p.f. 0.707 lagging.

(iii) Synchronous motor load of 50 kW at p.f. 0.9 leading.

Calculate the total kW and kVA delivered by the generator and the power factor at which it works.

b) A 2-wire d.c. ring distributor is 300 m long and is fed at 240 V at point A. At point B, 150 m from A, a load of 120 A is taken and at C, 100 m in the opposite 7 direction, a load of 80 A is taken. If the resistance per 100 m of single conductor is 0·03 Ω, find :

(i) current in each section of distributor

(ii) voltage at points B and C

20 a) A 800 metres 2-wire d.c. distributor AB fed from both ends is uniformly loaded at the rate of 1·25 A/metre run. Calculate the voltage at the feeding points A and B if the minimum potential of 220 V occurs at point C at a distance of 450 metres from the end A. Resistance of each conductor is 0·05 Ω/km. 6

b) An industry has a maximum load of 250kW at 0. 707p.f lag, with an annual consumption of 30,000units. The tariff is Rs.60/kVA of maximum demand plus 15 paise per unit. Calculate the following. (i) the flat rate of energy consumption. (ii) annual savings if the p.f is raised to unity. 8

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